Developing non-factualism about conditionals

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1. Recap: non-factualism in general; expressivism in particular

2. The "Frege-Geach problem" (or challenge)

The challenge: it's not enough to just talk about some family of simple sentences involving X-type vocabulary. We need a theory that gives us a systematic way of characterising the meanings of *all* well-formed sentences involving X-type vocabulary.

- This includes, e.g., negations and disjunctions, including 'mixed' ones. 'Lying is not wrong'; 'Either lying is wrong or gambling is wrong'; 'Either lying is wrong or snow is white'; 'Lying is wrong and gambling is wrong'.
- Also sentences with quantifiers and modals
- And mental state ascriptions. 'She believes that lying is wrong'; 'She is pretty confident that lying is wrong'; 'She hopes that lying is wrong'; 'She is pretty confident that either lying is wrong or gambling is wrong'…

Much of the history of discussion on this point consists in people saying very programmatic things — 'reasons for optimism' rather than theories.

- In the case of conditionals, there has considerable emphasis on the fact that *some* embeddings of conditionals are hard to interpret. OK, but many others are not and we need a theory for them

3. An obvious but problematic strategy (for an expressivist)

Associate operators ('not', 'and', 'or', 'necessarily', 'Joe believes that…') with *functions from mental states to mental states*. When $M(S_1) = m_1$ and $M(S_2)=m_2$, M('S_1 or S_2') = $f_\vee(m_1,m_2)$.

This looks super hard, even for factual sentences!

- What function do you apply to *believing snow is white* to get *believing that snow is not white*? (Note that this is not the same as *not believing that snow is white*, so the function isn't just the ordinary *negation* operation on properties.)
- What function do you apply to *believing snow is white* and *believing grass is green* to get back *believing that snow is white or grass is green*? (Note that this is not the same as *believing that snow is white or believing that grass is green*, so the function isn't just the ordinary disjunction operation on properties.

4. A more flexible, two-part strategy

Part 1: an assignment $[\![\cdot]\!]$ of *compositional semantic values.* $[\![S_1 \text{ or } S_2]\!] = [\![\text{or}]\!]([\![S_1]\!], [\![S_2]\!])$, etc.

Part 2: a *bridging principle* mapping these compositional semantic values onto mental states (if you're an expressivist), or more generally and ultimately, onto whatever the relations between sentences and linguistic communities the nonfactualist proposes to substitute for relations _ *expresses proposition P in* _.

- We can take the compositional semantic values to be whatever it's most convenient to take them to be, so long as we can formulate good bridging principles mapping these entities back to mental states (or whatever).
- The mapping doesn't have to be one-one. Perhaps there are two sentences that, when asserted on their own, would express the exact same mental state, but such that substituting one for another in some more complex sentence would change which mental state it expresses.

5. An example: trivalent semantics for expressivists

Semantic values of sentences are functions from the set of all possible worlds to the three-membered set {0, ½, 1}.

  - Semantic values of names are objects; semantic values of predicates are functions from name semantic values to sentence semantic values; semantic values of operators are functions from sentence semantic values to sentence semantic values; etc.
  - When S expresses a proposition P, (i.e. a factual sentence) $[\![S]\!]$ the "bivalent" function f such that $f(w)=1$ when the proposition is true at w and $f(w)=0$ otherwise. We assume that $[\![\text{and}]\!]$, $[\![\text{or}]\!]$, and $[\![\text{not}]\!]$ behave in the standard ways when given bivalent functions as arguments.

Bridging principle: a sentence with semantic value f expresses the mental state *having high credence in* $f^{-1}(\{1\})$ conditional on $f^{-1}(\{0, 1\})$.

- For example, suppose we have a sentence S whose semantic value is the function f where $f(w)=0$ if it is raining at w and the ground is not wet, $f(w)=½$ if it is not raining at w, and $f(w)=1$ if it is raining at w and the ground is wet. Then S expresses the mental state *having high conditional credence that the ground is wet given that it is raining*.

Some compositional rules:

$$[\![\text{If}]\!](f, g)(w) = \begin{cases} ½ \text{ if } f(w) = 0 \text{ or } f(w) = ½ \text{ or } g(w) = ½ \\ 0 \text{ if } f(w) = 1 \text{ and } g(w) = 0 \\ 1 \text{ if } f(w) = 1 \text{ and } g(w) = 1 \end{cases}$$

⟦Is at least n% confident that⟧(f)(x)(w)

= 1 if at w x's credence in $f^{-1}(\{1\})$ given $f^{-1}(\{0, 1\})$ is at least n%

= 0 otherwise

- *Consequence:* when A and B are sentences that express propositions P and Q, 'Mary is at least n% confident that if A, B' expresses a proposition that is true exactly those worlds where Mary's credence in Q given P is at least n%.
- Another interesting consequence: When A, B, C express propositions, ⟦If A, if B, C⟧ = ⟦If (A and B), C⟧ (here using the assumption that 'and' behaves standardly on bivalent arguments).

How should ⟦and⟧, ⟦or⟧, ⟦not⟧ work for non-bivalent arguments?

- For ⟦not⟧ there is only one reasonable choice, namely ⟦not⟧(f)(w) = 1–f(w). Otherwise we'd have a difference between ⟦A⟧ and ⟦not not A⟧.
- But for ⟦and⟧ and ⟦or⟧ there are several choices, the most natural of which are the Strong Kleene, Weak Kleene, and McDermott truth tables.

$\wedge_{wk}$	1	½	0
1	1	½	0
½	½	½	½
0	0	½	0

$\wedge_{sk}$	1	½	0
1	1	½	0
½	½	½	0
0	0	0	0

$\wedge_{McD}$	1	½	0
1	1	1	0
½	1	½	0
0	0	0	0

$\vee_{wk}$	1	½	0
1	1	½	1
½	½	½	½
0	1	½	0

$\vee_{sk}$	1	½	0
1	1	1	1
½	1	½	½
0	1	½	0

$\wedge_{McD}$	1	½	0
1	1	1	1
½	1	½	0
0	1	0	0

- Focusing on ⟦and⟧, we have the following rather surprising consequences, for factual sentences A, B, C, D:
  - If ⟦and⟧ is $\wedge_{wk}$, ⟦(If A, B) and C⟧ = ⟦If A, (B and C)⟧
    - In that case 'I am more confident that (if A, B) and C than that C' can be true (even if I'm rational)!
  - If ⟦and⟧ is $\wedge_{sk}$, ⟦(If A, B) and C⟧ = ⟦If (A or not C), (A and B and C)⟧
    - Not so clear if this is bad. But looks worse when we turn to conjunctions of two conditionals.

- If $[\![\text{and}]\!]$ is $\wedge_{McD}$, $[\![(\text{If A, B}) \text{ and C}]\!] = [\![(\text{not (A and not B)) and C}]\!]$
  - In that case 'I am more confident that (if A, B) and C than that if A, B' can be true!
- Turning to conjunctions of two conditionals, we have the following (exercise!):
  - If $[\![\text{and}]\!]$ is $\wedge_{wk}$, $[\![(\text{If A, B}) \text{ and } (\text{If C, D})]\!] = [\![\text{If (A and C), (B and D)}]\!]$
  - If $[\![\text{and}]\!]$ is $\wedge_{sk}$, $[\![(\text{If A, B}) \text{ and } (\text{If C, D})]\!] = [\![\text{If } A \wedge C \text{ or } A \wedge \neg B \text{ or } C \wedge \neg D, \text{ (A and B and C and D)}]\!]$.
  - If $[\![\text{and}]\!]$ is $\wedge_{McD}$, $[\![(\text{If A, B}) \text{ and } (\text{If C, D})]\!] = [\![\text{If (A or C), (B and D) or (A and D) or (B and C)}]\!]$.

To draw out the surprising consequence of the $\wedge_{sk}$ treatment, suppose we have a 100 ticket lottery. Consider the sentence $S_1 \wedge S_2$ = 'If a number under 51 is drawn a number under 50 will be drawn, and if a number over 49 is drawn a number over 50 will be drawn'. Its value is ½ (= $1 \wedge_{sk} ½$) at worlds where a number under 50 is drawn; 0 (=$0 \wedge_{sk} 0$) if 50 is drawn; and again ½ (= ½ $\wedge_{sk}$ 1) if the number drawn is great then 50. So it's equivalent to 'If 50 is drawn, 50 won't be drawn'. If you're rational, 'You are 0% confident that $S_1 \wedge S_2$' is true, despite the fact that (assuming you know the lottery is fair), 'You are 98% confident that S_1' and 'You are 98% confident that S_2' are both true. Curious!